

MindSpace

(Student Mental Health Support System)

A Web-Based Platform for Student Well-being, Counselling and Peer Support

Software Requirements Specification (SRS)

Field	Details
Project Name	MindSpace - Student Mental Health Support System
Document Type	Software Requirements Specification (SRS)
Student Name	D. D. Gajanayaka
Registration No.	DEH/IT/2324/F/0044
Institute	Sri Lanka Institute of Advanced Technological Education (SLIATE)
Programme	Higher National Diploma in Information Technology
Supervisor	Ms. H.A.P. Anusha
Date	July 2026

Contents

1. Introduction	4
1.1 Purpose	4
1.2 Scope	4
1.3 Definitions, Acronyms, and Abbreviations	5
1.4 References	5
2. Overall Description	6
2.1 Product Perspective	6
2.2 Product Functions	6
2.3 User Classes and Characteristics	7
2.4 Operating Environment	7
2.5 Constraints	8
2.6 Assumptions and Dependencies	8
3. Functional Requirements	9
3.1 User Registration and Authentication (FR1-FR3)	9
3.2 Mood Tracking and Gamification (FR4-FR5)	9
3.3 AI Chat Companion – Aura (FR6)	10
3.4 Counselling Booking System (FR7-FR8)	10
3.5 Counsellor Onboarding (FR9)	11
3.6 Anonymous Peer Support Chat (FR10)	11
3.7 Role-Based Dashboards (FR11-FR13)	12
3.8 Resources, Announcements and Feedback (FR14-FR16)	12
4. Non-Functional Requirements	13
4.1 Performance	13
4.2 Security	13
4.3 Usability	14
4.4 Reliability	14
5. External Interface Requirements	15
5.1 User Interface	15
5.2 Software Interfaces	15
6. System Constraints and Boundaries	16

6.1 In Scope	16
6.2 Out of Scope	16
7. Other Requirements	17
7.1 Data Requirements.....	17
7.2 Ethical Requirements	17
Appendix A: Glossary.....	18

1.Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the complete functional and non-functional requirements for MindSpace, a web-based student mental health support platform. It serves as the primary reference for the supervisor and evaluators throughout the design, implementation, and testing phases of the system.

1.2 Scope

MindSpace is a full-stack web application connecting students, counsellors, and administrators in a single platform dedicated to student well-being. The system aims to:

- Provide students with a private, judgement-free space to track and understand their daily mood and habits.
- Offer an always-available, rule-based bilingual (English/Sinhala) AI chat companion, “Aura”, for immediate emotional support.
- Enable students to book paid one-on-one counselling sessions with verified counsellors and track session status in real time.
- Allow verified professionals to apply, get vetted, and manage their own counselling caseload and earnings.
- Facilitate anonymous, real-time peer-to-peer support chat that never reveals a student's real identity.
- Give administrators tools to moderate users, verify counsellor applications, publish resources/announcements, and monitor platform activity.

*The system does NOT include real-time emergency/crisis medical response, clinical diagnosis, prescription management, or government health-system integration (out of scope for this release).

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification
MindSpace	The proposed student mental health support platform
JWT	JSON Web Token
API	Application Programming Interface
MERN	MongoDB, Express.js, React.js, Node.js technology stack
Aura	MindSpace's rule-based, bilingual AI chat companion
NIC	National Identity Card
UI/UX	User Interface / User Experience
LKR	Sri Lankan Rupee

1.4 References

- IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications
- React.js Documentation – <https://react.dev/>
- Node.js Documentation – <https://nodejs.org/docs>
- Express.js Documentation – <https://expressjs.com/>
- MongoDB & Mongoose Documentation – <https://www.mongodb.com/docs/> , <https://mongoosejs.com/docs/>
- Socket.IO Documentation – <https://socket.io/docs>

2. Overall Description

2.1 Product Perspective

MindSpace is an independently developed, standalone web application built on the MERN stack (MongoDB, Express.js, React.js, Node.js). It integrates with:

- Socket.IO for real-time notifications, live peer-chat matchmaking, and instant messaging
- MongoDB (via Mongoose ODM) for persistent storage of users, mood logs, bookings, payments, messages, and resources
- JSON Web Tokens (JWT) for stateless, token-based authentication and session management
- A built-in rule-based sentiment engine and the Aura chatbot, both implemented natively in the backend rather than via a third-party AI API

The system is designed to give school and university students continuous, low-barrier access to mental well-being tools that are not readily available through existing institutional counselling processes, and it does not depend on any existing legacy platform.

2.2 Product Functions

At a high level, MindSpace provides the following core functions:

- User registration, JWT-based login, and role-based access control (Student, Counsellor, Administrator)
- Daily mood tracking with rich context (sleep, hydration, screen time, exercise, energy level, voice/photo notes) and automatic streaks and badges
- AI-assisted sentiment analysis of mood journal entries with personalised coping suggestions
- “Aura” – an always-on, bilingual (English/Sinhala) rule-based chatbot companion
- Counselling session booking with live time-slot availability and simulated card payment
- Counsellor application, credential submission, and admin verification workflow
- Anonymous, alias-based real-time peer support chat with automatic matchmaking
- Role-based dashboards for Students, Counsellors, and Administrators
- Wellness resource library, platform announcements, and student feedback handling
- Counsellor payout tracking linked to completed, paid sessions

2.3 User Classes and Characteristics

User Class	Description	Access Level
Student	Logs mood, chats with Aura and peers anonymously, books and pays for counselling sessions, apply to become a Counsellor.	Manage own mood logs, bookings, and chat sessions; view own dashboard and earned badges.
Counsellor	A vetted professional who manages their availability, reviews and responds to booking requests, and tracks session payouts.	Approve/reject bookings; view own earnings and student notes for booked sessions.
Administrator	Oversees the platform verifies counsellor applications, manages users, publishes resources/announcements, and reviews feedback.	Full access: user management, counsellor verification, payout oversight, content publishing.

2.4 Operating Environment

- Web Browsers: Google Chrome, Microsoft Edge, Mozilla Firefox, Safari
- Client Devices: Windows, macOS, and mobile-responsive on Android/iOS browsers
- Frontend: React 19 single-page application bundled and served via Vite
- Backend: Node.js with Express 5, exposing a REST API and a Socket.IO real-time gateway
- Database: MongoDB (accessed through Mongoose schemas/models)
- Network: Broadband or mobile data connection with internet access

2.5 Constraints

- The AI chat companion and mood-sentiment engine are rule-based (keyword and metric driven); they are not a licensed medical or diagnostic AI service.
- Counselling session payment is simulated within the application (masked card details and a generated transaction ID are stored) – no real payment gateway is integrated in this version.
- Real-time features (peer chat, notifications) require an active Socket.IO connection; there is no offline mode.
- Counsellor availability is limited to what each verified counsellor manually configures.
- Base64 encoded images/audio (profile photos, ID documents, voice notes) are capped by a 10MB request body limit on the backend.

2.6 Assumptions and Dependencies

- Users have access to a smartphone or computer with a stable internet connection.
- Students and counsellors provide a valid, unique email address at registration.
- Counsellor applicants provide genuine identity and qualification documents for admin review.
- All users have basic digital literacy sufficient to use a web-based platform.
- The MongoDB database is hosted on infrastructure (e.g. a managed cloud database) accessible to the backend at all times.

3. Functional Requirements

3.1 User Registration and Authentication

FR1: User Registration

- The system shall allow new users to register with full name, email address, and password; the default role assigned is Student.
- The system shall validate the fact that email addresses are unique across all registered accounts.
- The system shall hash passwords using bcrypt before storing them; plaintext passwords never persisted.

FR2: User Login and Session Management

- The system shall authenticate users by comparing the submitted password against the stored bcrypt hash.
- The system shall issue a JSON Web Token (JWT), valid for 7 days, upon successful login.
- The system shall track each user's login count for activity/engagement reporting.
- The system shall reject login for accounts whose status has been set to 'deactivated' by an Administrator.

FR3: Role-Based Access Control

- The system shall support three user roles: Student, Counsellor, and Administrator, each with a distinct dashboard and permission set.
- The system shall allow Administrators to activate or deactivate any user account.
- The system shall verify the bearer JWT on every protected API request and reject unauthenticated or invalid requests.

3.2 Mood Tracking and Gamification

FR4: Daily Mood Logging

- The system shall allow Students to log a mood entry consisting of an emoji, mood label, and numeric mood value.
- The system shall allow an optional journal note, a trigger tag, and a list of activities associated with the entry.
- The system shall allow logging of supporting wellness metrics: sleep hours, water intake, screen time, exercise duration, and energy level.
- The system shall allow an optional voice note and photo to be attached to a mood entry.
- The system shall allow Students to record contextual tags such as weather, music genre, and whether the entry falls within an exam period.
- The system shall display mood history as interactive charts (via Recharts) so Students can review trends over time.
- The system shall allow Students to edit or delete their own mood entries.

FR5: Streaks, Badges, and AI Sentiment Feedback

- The system shall calculate a consecutive-day logging streak each time a Student submits a mood entry.
- The system shall automatically award badges based on behaviour, including “3-Day Explorer”, “7-Day Streak Warrior”, “30-Day Zen Master”, “Rest & Recover” (8+ hours sleep), “Hydration Hero” (2000ml+ water), and “Active Soul” (3+ activities logged).
- The system shall run a rule-based sentiment analysis over each mood journal entry — considering the note text alongside sleep, hydration, screen time, and exam-period context — to produce a sentiment score, label, and a short list of coping suggestions.
- The system shall generate personalized recommendations (e.g. a suggested game or activity) stored against the student’s profile.

3.3 AI Chat Companion - Aura

FR6: Rule-Based Bilingual AI Companion

- The system shall provide an in-app chatbot named “Aura” that responds instantly to Student messages using a built-in rule-based response engine (not an external AI API).
- The system shall detect emotional categories from user input such as greetings, sadness, anxiety, stress, and loneliness and reply with a contextually appropriate, empathetic message.
- The system shall support responses in both English and Sinhala.
- The system shall offer light-touch coping techniques within the conversation, such as guided breathing prompts, without providing medical diagnosis or treatment.

3.4 Counselling Booking System

FR7: Session Booking and Scheduling

- The system shall allow Students to view the list of verified, active Counsellors and their available time slots.
- The system shall allow Students to submit a booking request for a specific date and time slot, with an optional note.
- The system shall allow Counsellors to approve or reject a booking request, optionally providing a rejection reason.
- The system shall support booking statuses of pending, approved, rejected, and cancelled.

FR8: Simulated Session Payment

- The system shall require Students to complete a simulated card payment (card holder name and masked card number) to confirm an approved booking.
- The system shall generate a transaction ID and paid timestamp for each simulated payment and mark the booking's payment status as ‘paid’.
- The system shall create a corresponding Payment record (default session rate LKR 1,500) linking the Student, Counsellor, and Booking for payout tracking.
- The system shall allow Counsellors to view their own earnings and Administrators to view all counsellor payouts, including counsellor bank account details on file.

3.5 Counsellor Onboarding

FR9: Counsellor Application and Verification

- The system shall allow a registered user to apply to become a Counsellor by submitting full name, NIC number, profile photo, email, phone number, qualifications, licence number, professional membership, and front/back images of their counsellor ID.
- The system shall require applicants to declare years of experience, prior institutions, and areas of specialization.
- The system shall set new applications to a 'pending' status and allow Administrators to approve or reject each application.
- The system shall upgrade an approved applicant's account role to Counsellor upon approval.

3.6 Anonymous Peer Support Chat

FR10: Real-Time Anonymous Matchmaking Chat

- The system shall allow a Student to join a peer-support matchmaking queue under a randomly generated alias (e.g. "Anonymous Hippo") rather than their real name.
- The system shall automatically pair a waiting Student with another different Student from the queue and open a private, real-time chat room between them using Socket.IO.
- The system shall never expose either participant's real identity within the peer-chat room.
- The system shall persist chat messages against the anonymous room identifier for moderation and history purposes.
- The system shall notify both participants in real time if the other party disconnects or leaves the chat and allow either party to leave the queue or the chat at any time.

3.7 Role-Based Dashboards

FR11: Student Dashboard

- The system shall display a Student dashboard summarizing recent mood trends, current streak and earned badges, upcoming bookings, and unread notifications.

FR12: Counsellor Dashboard

- The system shall display a Counsellor dashboard listing incoming booking requests, confirmed sessions, and payout/earnings history.

FR13: Administrator Dashboard

- The system shall display an Admin dashboard summarizing registered users, pending counsellor applications, booking activity, and outstanding feedback.
- The system shall allow Administrators to activate, deactivate, or review any user account.
- The system shall allow Administrators to publish platform-wide or role-targeted announcements.

3.8 Resources, Announcements and Feedback

FR14: Wellness Resource Library

- The system shall allow Administrators to publish wellness articles categorised as Stress, Anxiety, Sleep, or Mindfulness, each with an estimated read time and author attribution.
- The system shall support resources in both English and Sinhala.

FR15: Announcements

- The system shall allow Administrators to broadcast announcements targeted at all users, Students only, or Counsellors only.

FR16: Feedback System

- The system shall allow Students to submit feedback with a subject and message.
- The system shall allow Administrators to track feedback status as pending, resolved, or done.

4. Non-Functional Requirements

4.1 Performance

- The system shall load dashboard and marketplace-style listing pages within 3 seconds under normal operating conditions.
- The Aura chatbot shall return a response within 2 seconds of a Student message, since responses are generated locally without an external API round-trip.
- Real-time events (peer-chat matching, messages, notifications) delivered via Socket.IO shall reach connected clients within 2 seconds of the triggering event.
- The system shall support at least 500 concurrent users without noticeable degradation.

4.2 Security

- User passwords shall be hashed using bcrypt before storage; plaintext passwords shall never be stored.
- All protected API endpoints shall require a valid JWT in the Authorization header; unauthenticated or invalid requests shall return HTTP 401.
- Deactivated accounts shall be denied access at both login and on every subsequent authenticated request.
- Sensitive counsellor identity documents and bank details shall only be viewable by Administrators and the owning Counsellor.
- Anonymous peer-chat participants shall never have their real name, email, or user ID exposed to their chat partner.
- All traffic between client and server shall be served over HTTPS in production.

4.3 Usability

- The user interface shall be fully responsive, usable from mobile screens (360px) up to desktop displays (1920px).
- Core student journeys logging a mood, starting an Aura chat, or booking a counselling session shall be completable within 5 clicks from the dashboard.
- The interface shall support English and Sinhala for chatbot responses and wellness resources, with the architecture allowing further language expansion.
- The system shall display clear inline validation messages on all form fields.

4.4 Reliability

- The system shall target 99% uptime for the web application and API.
- The MongoDB database shall be backed up on a regular schedule to prevent data loss.
- The system shall log server-side errors for monitoring and troubleshooting.
- The system shall handle a dropped Socket.IO connection gracefully, notifying the affected peer-chat partner and cleaning up the matchmaking queue.

5. External Interface Requirements

5.1 User Interface

The UI shall be built with React (Vite) and plain CSS. Key design principles:

- A persistent, role-aware navigation bar (Navbar) linking to Mood Tracker, Booking, AI Chatbot, Anonymous Chat, Resources, and the relevant dashboard.
- Interactive mood-trend charts rendered with Recharts.
- A conversational chat interface for both the Aura chatbot and the anonymous peer chat.
- A booking interface showing counsellor availability by date and time slot.
- Badge and streak indicators to visually reinforce positive logging habits.

5.2 Software Interfaces

External System	Interface Type	Purpose
Socket.IO	WebSocket	Real-time peer-chat matchmaking, messaging, and live notifications
MongoDB (Mongoose)	TCP / ODM	Persistent storage for users, mood logs, bookings, payments, messages, resources
JWT	HTTP Header (Bearer token)	Stateless authentication and authorisation for all protected API routes
REST API (Express)	HTTPS/JSON	Core application logic for auth, mood, bookings, payments, resources, and admin operations

6. System Constraints and Boundaries

6.1 In Scope

- User registration and role-based authentication (Student, Counsellor, Administrator).
- Daily mood tracking with wellness metrics, streaks, badges, and AI-generated sentiment feedback.
- Rule-based bilingual AI chat companion (Aura) for immediate emotional support.
- Counselling session booking with simulated payment and counsellor payout tracking.
- Counsellor application and admin verification workflow.
- Anonymous, real-time peer support chat with automatic matchmaking.
- Role-based dashboards for Students, Counsellors, and Administrators.
- Wellness resource library, announcements, and student feedback handling.

6.2 Out of Scope

- Real-time emergency or crisis medical response services.
- Clinical diagnosis, prescription, or medication management.
- Integration with a real, third-party payment gateway (payment is simulated in this version).
- Government or national health-system integration.
- Native iOS or Android mobile applications.
- Use of a third-party generative-AI provider for the chatbot or sentiment engine.

7. Other Requirements

7.1 Data Requirements

- User profiles, including role, status, gamification data (streaks, badges, monthly goal logs), and counsellor bank details where applicable.
- Mood log records with all associated wellness metrics and AI sentiment results.
- Booking and payment records linking Students, Counsellors, and session payouts.
- Counsellor application records, including submitted credentials and verification status.
- Anonymous peer-chat and Aura conversation message history, keyed by chat room.
- Wellness resources, announcements, and feedback records.

7.2 Ethical Requirements

- Student privacy and confidentiality shall be always protected, particularly for mood journal content and chat history.
- The anonymous peer-chat feature shall ensure no participant's real identity can be inferred from the alias or displayed information.
- The Aura chatbot and sentiment engine shall never present themselves as a substitute for a licensed mental health professional and shall avoid giving harmful or medically prescriptive content.
- Counsellor's identity and financial information shall be handled with restricted, role-based access.

Appendix A: Glossary

Term	Definition
Aura	MindSpace's built-in, rule-based bilingual chatbot that offers immediate emotional support to Students.
Mood Log	A Student's daily record of mood, journal note, and wellness metrics (sleep, hydration, screen time, exercise, energy).
Streak	The number of consecutive days a student has logged a mood entry.
Badge	A gamification reward automatically granted for reaching a wellness or engagement milestone (e.g. “7-Day Streak Warrior”).
Sentiment Analysis	The system's rule-based evaluation of a mood journal entry to derive a sentiment score, label, and coping suggestions.
Simulated Payment	An in-app, mock card-payment flow used to mark a counselling booking as paid without connecting to a real payment gateway.
Counsellor Application	The vetting workflow through which a registered user submits credentials to be verified and promoted to the Counsellor role.
Anonymous Peer Chat	A real-time, alias-based chat feature that matches two Students for support conversation without revealing their identities.
JWT (JSON Web Token)	A compact, URL-safe token used to securely transmit user identity and claims between the client and server.
Role-Based Access Control	A security model that restricts system access and functionality based on a user's assigned role.